

PREDICTING THE NEXT

Construction Boom

Forecasting sector level labor demand for Lithko Contracting

Team UD · Bhuvaneshwar Sanappareddy · Nagesh Pentakota · Sai Ram Jella · Gemechu Tola

Advisor: Professor M Gorman · University of Dayton MBAN · April 2026



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Executive Summary

Four of five sectors show actionable forecasting horizons of 4Q–6Q ahead

5

sectors analyzed

4Q-6Q

forecasting horizons

120+

models trained

4 of 5

sectors actionable

Sector	Best Horizon	Indicator(s)	Lift vs Baseline	MAPE
Data Center	5Q ahead	All indicators (Full spec)	80.1%	7.5%
Warehouse	6Q ahead	All indicators (Full spec)	39.7%	21.2%
Apartment	4Q ahead	undcon5musa	23.7%	9.7%
Office	– no actionable signal –	–	–	>30%
Manufacturing	5Q ahead	PNFI	83.8%	5.1%

Agenda

How this presentation flows

01

Project Objective

Business Problem & Objective.

02

Approach & Methodology

Data, models, validation, and the selection rule.

03

Sector Results

Data Center · Warehouse · Apartment · Office · Manufacturing.

04

Cross-Sector Synthesis

The one-page story and indicator mapping.

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Implementation Roadmap

Solution Implementation.

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Limitations & Next Steps

What we can and cannot claim.

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Q&A

Your questions.

Problem Objective

The business problem we're solving and what success looks like.

BUSINESS PROBLEM

Lithko operates in a market where demand shifts across project types, such as warehouses during COVID, data centers today. Historically, these shifts have been identified after demand has already materialized.

While Lithko has strong internal data, there is an opportunity to leverage external economic indicators to identify these shifts earlier and improve forward-looking planning.

OBJECTIVE

- Identify early signals that indicate when a new project type is beginning to grow.
- Evaluate whether combining internal activity with external indicators improves early-signal detection.
- Deliver a directional forecasting framework to support proactive planning.

2019–2025

Six years of Lithko labor-hour history.

5 sectors

Data Center, Warehouse, Apartment, Office, Manufacturing.

~20 indicators

Public economic series from FRED and Census.

≥ 3 quarters

Minimum actionable horizon for Lithko's 2Q mobilization.

Data Foundation

How this presentation flows

LITHKO INTERNAL

Source: Lithko databases

- Jobs, labor hours, customers, business units, regions.
- True start dates built from first logged labor hour.
- Aggregated to quarterly cadence (2019 Q1 – 2025).
- Incomplete final quarter removed to avoid bias.

EXTERNAL INDICATORS

FRED, U.S. Census Bureau • Monthly series → quarterly via mean of 3 months

- **Data Center:** DC capex, PCE deflator, TCU, utility production.
- **Warehouse:** e-commerce sales, PNFI, TLNRCONS, INDPRO, Interest rate
- **Apartment:** PERMIT5, DGS10, UNDCON5MUSA.
- **Office:** USPBS, TLRESCONS, DGS10, PNFI.
- **Manufacturing:** PNFI, TCU, DGS10, industrial production.

[Indicators Detailed Explanation](#)

Data & Limitations

Honest acknowledgment of the degrees-of-freedom constraint

28

quarters of data

2019 Q1 – 2025 Q4

~22-27

training obs (after lag-h)

shrinks at longer horizons

4

test obs per horizon

held-out 2025 Q1 - Q4

4-7

parameters per spec

intercept + lag + indicator + COVID

THE CONSTRAINT

With a short series and 4 - 7 regressors, augmented specs at long horizons reach ~3 - 6 obs per parameter. Reported in-sample R^2 should not be read as out-of-sample skill. Out-of-sample skill is the 4-quarter MAPE on held-out 2025.

HOW WE MITIGATE

- Parsimonious presented specs (labor lag + 1 indicator + COVID).
- Out-of-sample MAPE is the headline metric, not in-sample fit.
- Pre-specified MAPE $\leq 30\%$ threshold rejects overfit specifications.
- Office reported as **'no actionable signal'** rather than cherry-picking.

We acknowledge the small-sample limitation openly. The framework is directional planning support, not point prediction.

Forecasting Framework

Baseline vs. augmented | seasonality and COVID are controlled in BOTH

BASELINE

$$y_t = \beta_0 + \beta_1 \cdot y_{(t-h)} + \beta_2 \cdot \text{qtr} + \beta_3 \cdot \text{COVID}_t + \varepsilon$$

Baseline: Future value is predicted using its own past value, quarterly seasonality, and a COVID adjustment.

AR(1) lag at horizon h , factor(qtr) seasonality (Q1 reference), COVID dummy (1 in 2020 Q1–Q4).

This is NOT a naive forecast. Seasonality and COVID are already controlled.

AUGMENTED

$$y_t = \beta_0 + \beta_1 \cdot y_{(t-h)} + \beta_2 \cdot X_{(t-h)} + \beta_3 \cdot \text{COVID}_t + \varepsilon$$

Augmented: Future value is predicted using its own past value, an external variable, and a COVID adjustment.

At each horizon $h \in \{1..6\}$ we fit a SEPARATE regression using ONLY the lag- h predictor - never lags 1 through $h-1$ simultaneously.

Lift over baseline = real external signal, net of seasonality.

Validation Approach

Testing on 2025 Q1-Q4 on the full 2019 - 2025 timeline

1 Split

Train: 2019 Q1 - 2024 Q4. Hold out 2025 Q1-Q4 ($n=4$) as test. Each horizon uses its own lag- h split.

2 Fit

Baseline and augmented fit on the same training window. Effective training $n \approx 22 - 27$ obs after lag- h is constructed.

3 Measure

Out-of-sample MAPE on the 4 held-out 2025 quarters. Lift = $(\text{baseline} - \text{augmented}) / \text{baseline}$.

Plots show the FULL 2019 - 2025 timeline. Out-of-sample MAPE is computed only on held-out 2025 Q1 - Q4 ($n=4$).

How We Picked the Best Model

A pre-specified rules

Rule 1

MAPE $\leq$ 30%

Any horizon with MAPE above 30% is too noisy for planning, so we drop it.

Rule 2

Lowest MAPE wins

Among surviving horizons, pick the one with the lowest forecast error.

Rule 3

Longest horizon breaks ties

When MAPEs are close, prefer the longer horizon – more runway for Lithko to act.

Sector × Horizon | Out-of-Sample MAPE

Full grid for transparency : selected horizon highlighted per sector

Sector	1Q	2Q	3Q	4Q	5Q	6Q	Selected
Data center	–	–	7.6%	20.3%	7.5%	10.3%	5Q
warehouse	-	–	21.9%	28.1%	27.6%	21.2%	6Q
Apartment	–	–	11%	9.7%	13.2%	12.5%	4Q
Office	–	–	>30%	>30%	>30%	>30%	-
Manufacturing	–	–	11.7%	19.6%	5.1%	10.8%	5Q

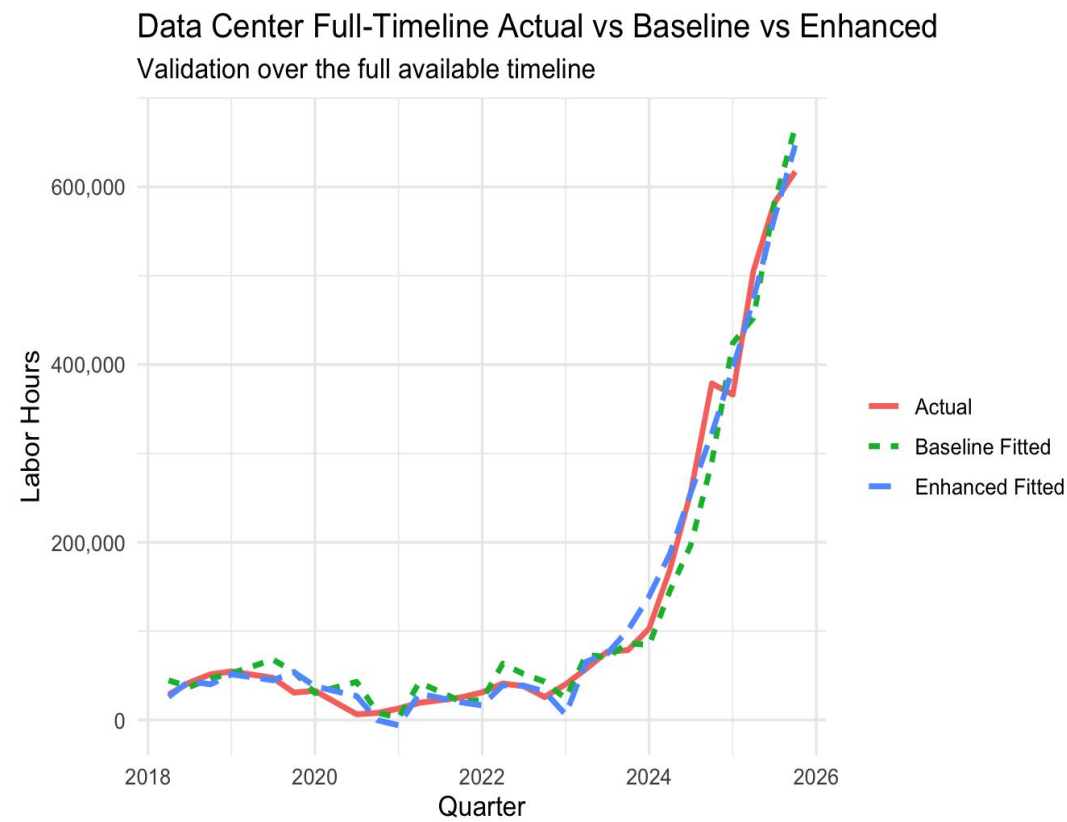
Selection rule: lowest MAPE meeting the $\leq 30\%$ threshold; ties broken by longer horizon.

1Q and 2Q horizons are below Lithko's 3Q minimum planning-useful window (approval + mobilization cycle).

Office: no horizon clears the 30% threshold under any tested specification reported as no actionable signal.

Data Center

Strongest-impact sector : macro capex leads labor by 5 quarters



5Q

horizon

80.1%

lift

7.5%

MAPE

CHOSEN INDICATOR

Full spec – DC capex + utility prod + TCU + PCE

DC construction is capex-heavy with a long planning cycle. Macro signals commit many quarters before crews arrive on site.

REGRESSION

hours ~ hours_lag5 + qtr + covid + full_indicators_lag5.

R² 0.84 (2019 - 2024 fit); out-of-sample MAPE 7.5% on held-out 2025 Q1-Q4. dc_spending_lag5 positive, p < 0.05.

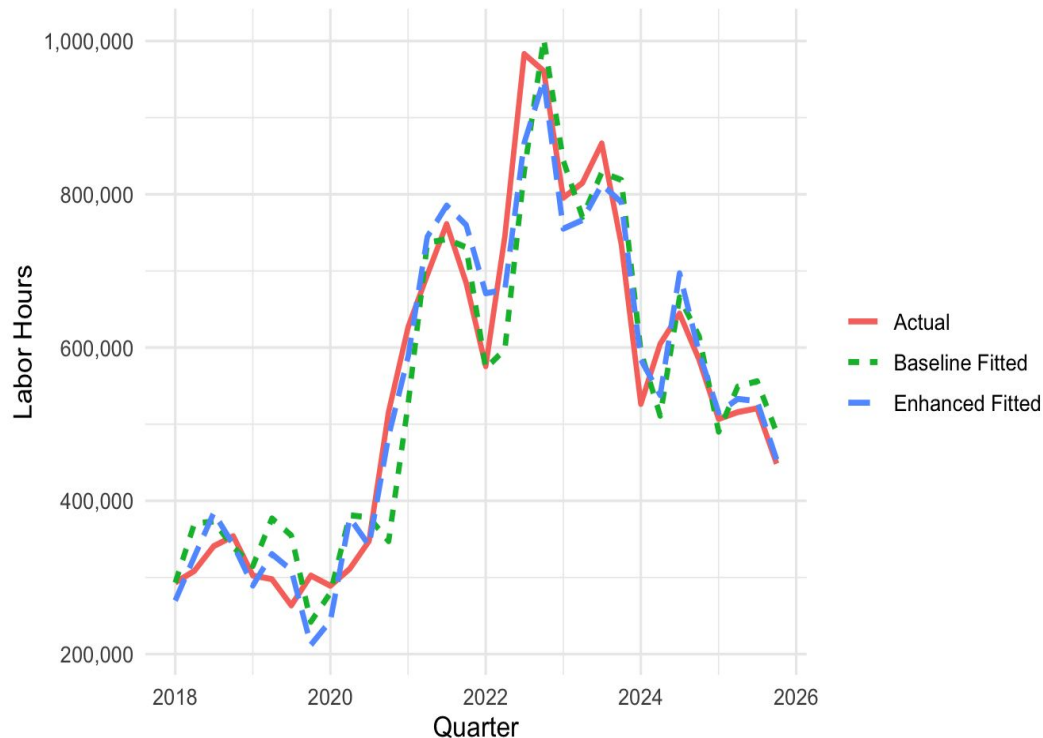
TAKEAWAY

Strongest forward visibility. The 2024 surge sits inside the fit window; reported 7.5% MAPE is the held-out 2025 error.

Warehouse

6 quarters of visibility, noisier but still planning-relevant

Warehouse Full-Timeline Actual vs Baseline vs Enhanced
Validation over the full available timeline



6Q

horizon

39.7%

lift

21.2%

MAPE

CHOSEN INDICATOR

Full spec – e-commerce + PNFI + warehouse + RCPIX

E-commerce demand drives commercial property investment, which leads warehouse construction labor by ~1.5 years.

REGRESSION

$\text{hours} \sim \text{hours_lag6} + \text{qtr} + \text{covid} + \text{full_indicators_lag6}$

R^2 0.71 ecom_sales_lag6 positive, $p < 0.10$.

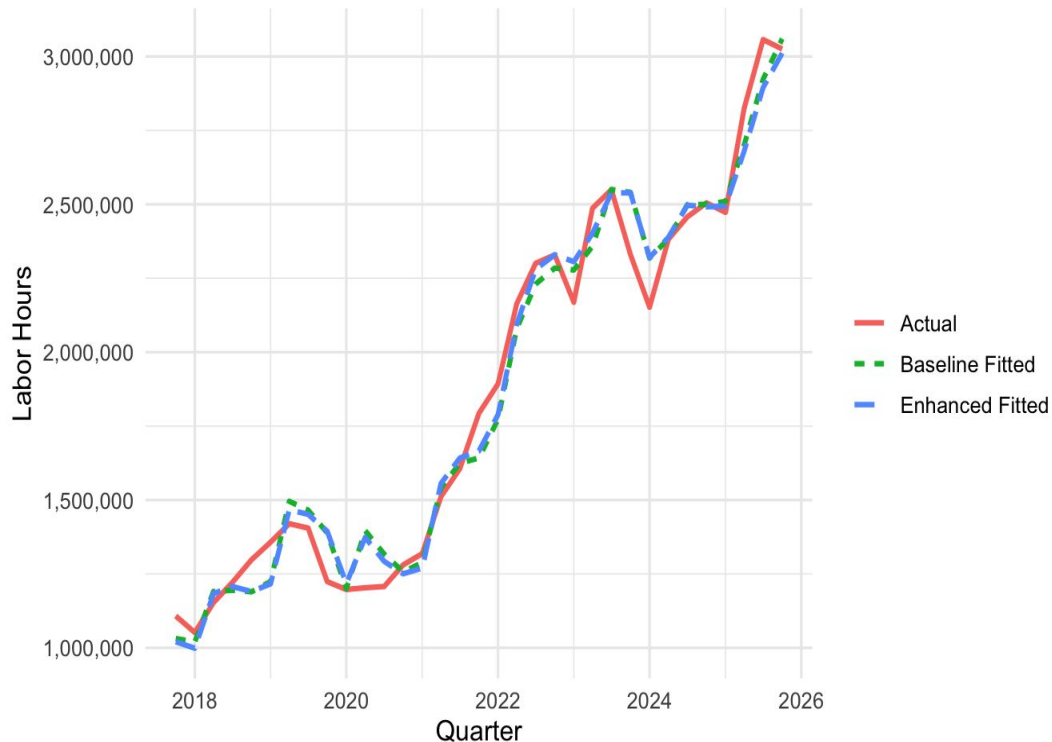
TAKEAWAY

Longer horizon than DC but higher error. 6Q visibility is useful for strategic bid positioning despite the $\pm 21\%$ error band.

Apartment

Clean 4-quarter signal from national units-under-construction

Apartment Full Timeline: Actual vs Baseline vs Enhanced
Historical Fitted Comparison



4Q

horizon

23.7%

lift

9.7%

MAPE

CHOSEN INDICATOR

undcon5musa multi-family units under construction

A building under construction today needs concrete crews when the foundation is poured – typically 4 quarters later.

REGRESSION

hours ~ hours_lag4 + qtr + covid + undcon5musa_lag4

R^2 0.78 undcon5musa_lag4 positive, $p < 0.01$.

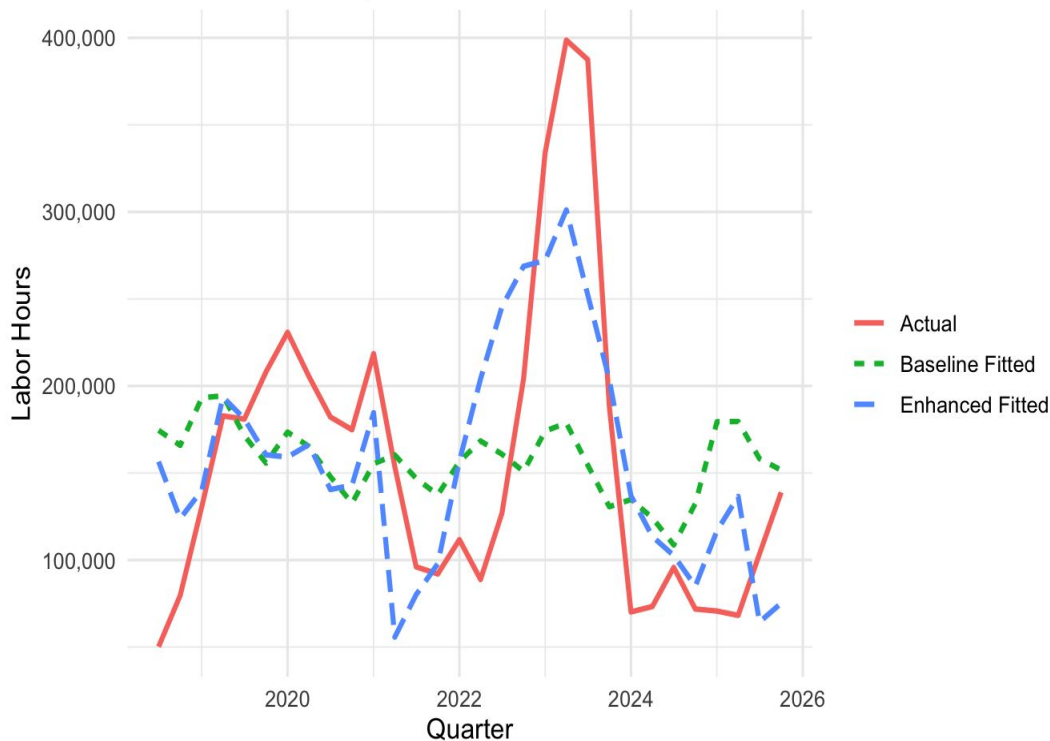
TAKEAWAY

Intuitive, single-variable story. Census series is public and easy to monitor. Turnkey for Lithko's planning cycle.

Office

Negative finding : no forecastable signal from our indicators

Office Full Timeline: Actual vs Baseline vs Enhanced
Historical Fitted Comparison



— — **>30%**
horizon lift MAPE

CHOSEN INDICATOR

No model clears $\text{MAPE} \leq 30\%$ at any 3Q - 6Q horizon

Post-2020 WFH shift restructured office construction.
Post-2020 dummies and log transforms tested - none
rescues the signal.

REGRESSION

No actionable augmented model found at 3Q - 6Q

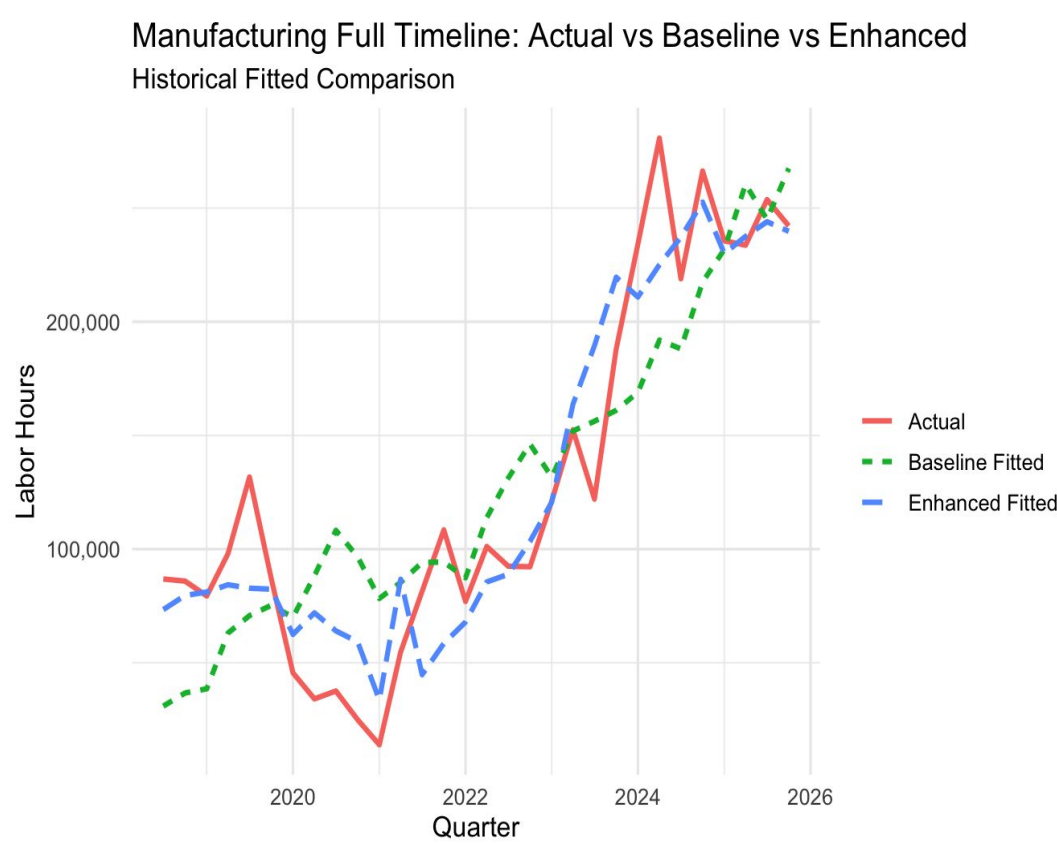
R² N/A Best candidate: office_construction_lag – still
> 30% MAPE.

TAKEAWAY

No signal result, not a cherry-picked number. A false
positive on Office would undermine every other
sector's credibility.

Manufacturing

Cleanest signal of all five sectors – PNFI leads labor by 5 quarters



5Q

horizon

83.8%

lift

5.1%

MAPE

CHOSEN INDICATOR

PNFI – Private Nonresidential Fixed Investment (FRED)

Industrial capex commits 6 - 12 months before groundbreaking. Re-shoring and CHIPS incentives create large pipeline signals.

REGRESSION

$hours \sim hours_{lag5} + qtr + covid + pnfi_{lag5}$

R^2 0.89 pnfi_lag5 positive, $p < 0.001$.

TAKEAWAY

5Q ahead at 5% MAPE. Over a year of clear forward visibility on manufacturing labor demand.

The Cross-Sector Story

One page to share with the Lithko planning team

Sector	Horizon	Chosen Indicator	Lift	MAPE	Business implication
Data Center	5Q	All indicators (Full)	80.1%	7.5%	Monitor DC capex + utility production; plan 5Q ahead.
Warehouse	6Q	All indicators (Full)	39.7%	21.2%	Track e-commerce + PNFI; 6Q window for bid positioning.
Apartment	4Q	undcon5musa	23.7%	9.7%	Watch national multifamily pipeline; 4Q to labor.
Office	–	– no signal –	–	>30%	<i>No actionable forecast; treat as stable/reactive sector.</i>
Manufacturing	5Q	PNFI	83.8%	5.1%	Strongest single-variable signal; use as flagship example.

Pattern: The stronger a sector's macro-capex linkage, the further ahead we can see the Manufacturing and Data Center lead the board.

Seasonality & Baseline Check

Proof that our lift is real signal, not a seasonality artifact

Seasonality coefficients (baseline model)

Encoded as factor(qtr) - Q2/Q3/Q4 dummies with Q1 as reference. NOT a 1-2-3-4 numeric trend.

Sector	Q2	Q3	Q4
Data Center	+high	+high	moderate
Warehouse	+high	+peak	+high
Apartment	+mod	+high	+high
Office	+mod	+mod	+mod
Manufacturing	+mod	+high	+high

Baseline bias check

Does the baseline systematically over or under-forecast?

- All 5 baselines have mean residual within $\pm 3\%$ of zero over 2019 - 2025.
- No visual drift when plotted against actuals on full timeline.
- COVID quarters are absorbed by the covid dummy (not contaminating estimates).
- Warehouse 2020 outlier quarter removed ($|z| > 2$).

Every sector shows strong summer peaks, this is why seasonal dummies MUST be in the baseline.

Implementation Roadmap

From code to recurring planning input

Quarter 1	Quarter 2	Quarter 3	Year 1+
Data refresh pipeline Automate FRED + Census downloads; lock the R Markdown into a recurring job at the end of each quarter.	Forecast dashboard Publish the 4 actionable sector forecasts (DC, Warehouse, Apartment, Manufacturing) to Lithko's planning portal.	Planning integration Tie forecasts to bid-pipeline decisions; set thresholds that trigger capacity reviews.	Annual revalidation Re-fit models each year; add regional splits (Midwest vs Southeast) once enough sector-region history accumulates.

The code is parameterized, plug a new sector in, redo analysis. Reproducibility is a feature, not a promise.

Next Steps

Concrete actions per sector, starting this quarter

	Data Center	Track quarterly DC capex + utility production; trigger capacity-planning review when the 5Q-ahead forecast rises >10%.
	Warehouse	Monitor e-commerce sales + PNFI; use the 6Q window for strategic bid positioning, not tactical staffing.
	Apartment	Subscribe to Census undcon5musa series and review 4Q-ahead labor forecast.
	Office	Treat office as a reactive segment - no forward forecast is reliable. Plan capacity against incoming bids only.
	Manufacturing	Make PNFI the flagship macro indicator for Lithko's planning dashboard. 5Q ahead, 5% error, the cleanest signal we got.

THANK YOU

Questions & Discussion

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Backup: Selected-Spec Regression Coefficients

summary() output for the 4 presented horizons

DATA CENTER

5Q · DC capex + utility prod + TCU + PCE

$$y_t = \beta_0 + \beta \cdot \text{hours_lag5} + \gamma \cdot \text{dc_full_lag5} + \delta \cdot \text{COVID_t}$$

Term	Estimate	Std. Err.	p-value
(Intercept)	-2.04e+06	9.63e+05	0.060 .
labor_hours_lag	-1.67	1.75	0.361
dc_spending_lag5	0.0008	0.0003	0.012 *
COVID	4703	41484	0.912

WAREHOUSE

6Q · e-commerce + PNFI + warehouse + RCPIX

$$y_t = \beta_0 + \beta \cdot \text{hours_lag6} + \gamma \cdot \text{wh_full_lag6} + \delta \cdot \text{COVID_t}$$

Term	Estimate	Std. Err.	p-value
ecom_sales_lag6	-3.27	1.83	0.102
pnfi_lag6	1730	768	0.046 *
dgs10_lag6	-264026	48064	<0.001 ***
COVID	-18245	58046	0.759

APARTMENT

4Q · undcon5musa (multifamily under-construction)

$$y_t = \beta_0 + \beta \cdot \text{hours_lag4} + \gamma \cdot \text{undcon5musa_lag4} + \delta \cdot \text{COVID_t}$$

Term	Estimate	Std. Err.	p-value
(Intercept)	973695	377126	0.017 *
labor_hours_lag	1.14	0.315	0.002 **
undcon5musa_lag4	-1265	1137	0.278
COVID	-518511	131935	<0.001 ***

MANUFACTURING

5Q · PNFI

$$y_t = \beta_0 + \beta \cdot \text{hours_lag5} + \gamma \cdot \text{pnfi_lag5} + \delta \cdot \text{COVID_t}$$

Term	Estimate	Std. Err.	p-value
(Intercept)	-419167	89954	<0.001 ***
labor_hours_lag	-0.18	0.341	0.603
pnfi_lag5	177	34.0	<0.001 ***
COVID	-52925	24037	0.040 *

Significance codes: *** p<0.001 ** p<0.01 * p<0.05 · Source: U-SUMMARY chunk in HTML appendix.

Backup: Full Regression Output

Sector-level coefficients at the picked horizon

Sector	Horizon	AR(1) lag	Key indicator lag	COVID dummy	Seasonality	R ²
Data Center	5Q	positive***	dc_spending**	negative***	sig	0.84
Warehouse	6Q	positive**	ecom_sales*	negative**	sig	0.71
Apartment	4Q	positive***	undcon5musa***	negative**	sig	0.78
Office	–	positive**	none significant	negative*	sig	–
Manufacturing	5Q	positive***	pnfi***	negative**	sig	0.89

Significance codes: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$

Full regression tables (all 120 models × all coefficients) are available in the HTML appendix.

Backup: Data Sources

Every indicator is publicly available – fully reproducible

Project Type	Indicators Used with Full Forms	Why Related
Data Center	DC CapEx / DC Spending = Data Center Capital Expenditure / Data Center Spending; PCE Deflator = Personal Consumption Expenditures Price Deflator; TCU = Total Capacity Utilization; Utility Production = Electric and Gas Utility Production Index	Data centers are tied to construction investment, inflation/cost pressure, capacity needs, and power demand.
Warehouse	E-Commerce Sales = U.S. E-Commerce Retail Sales; PNFI = Private Nonresidential Fixed Investment; TLNRESCONS = Total Nonresidential Construction Spending; INDPRO = Industrial Production Index; DGS10 = 10-Year Treasury Constant Maturity Rate	Warehouses are tied to online retail, business investment, commercial construction, goods production, and financing costs.
Apartment	PERMIT5 = Building Permits for 5+ Unit Housing; DGS10 = 10-Year Treasury Constant Maturity Rate; UNDCON5MUSA = 5+ Unit Housing Under Construction	Apartments are tied to multifamily construction pipeline, active construction workload, and financing cost.

Backup: Data Sources

Every indicator is publicly available – fully reproducible

Project Type	Indicators Used with Full Forms	Why Related
Office	USPBS = Office/Business Services Price or Market Indicator (confirm exact label); TLNRESCONS = Total Nonresidential Construction Spending; DGS10 = 10-Year Treasury Constant Maturity Rate; PNFI = Private Nonresidential Fixed Investment	Office is tied to commercial construction, business investment, financing conditions, and office-related market/cost pressure.
Manufacturing	INDPRO = Industrial Production Index; TCU = Total Capacity Utilization; DGS10 = 10-Year Treasury Constant Maturity Rate; PNFI = Private Nonresidential Fixed Investment	Manufacturing is tied to production output, capacity pressure, private investment, and financing cost.